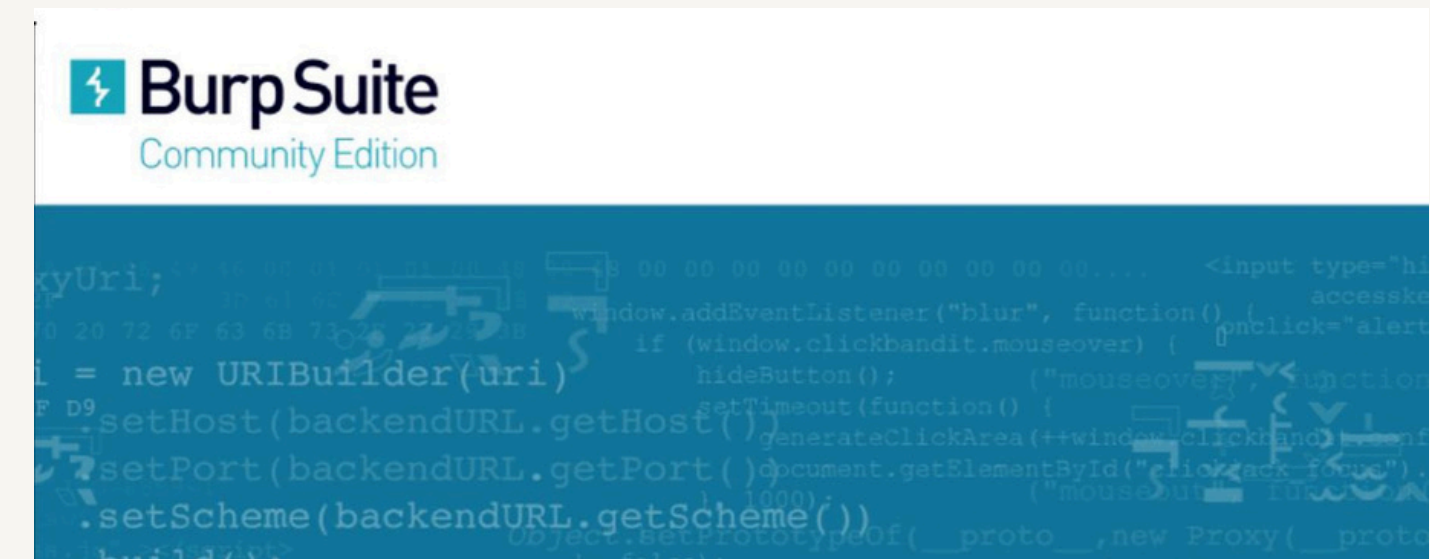




WEB APPLICATION SECURITY TESTING USING BURP SUITE

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This project demonstrates the practical use of Burp Suite for web application security testing. The project focuses on intercepting web traffic, analyzing requests and responses, identifying security weaknesses, and understanding the penetration testing workflow.



Introduction

what is Web Application Security ?

Web applications are used in banking, shopping, education, healthcare, and social media. These applications often store sensitive user information

If security controls are weak, attackers may:

- steal user data
- Hijack accounts
- Access confidential information
- Damage business reputation

why security Testing is important?

Security testing helps organization identify Vulnerabilities before attacker exploit them.

Project Objective

Learning Objectives

- Understand Burp Suite functionality
- Learn HTTP and HTTPS traffic analysis
- Capture browser requests
- Analyze application behavior
- Identify common security weaknesses

Practical Objectives

- Configure Burp Suite
- Intercept requests
- Modify request
- Analyze responses
- Generate findings

What is Burp Suite?

Burp suite is a web application security testing platform developed by portswigger.

It acts as as intermediary between the brower and the web application

Features

- Proxy
- Target
- Repeater
- Intruder
- Decoder
- Comparer

Benefits

- Traffic monitoring
- Request manipulation
- Vulnerability testing
- Security assessment

System Requirement

Hardware Requirement

Processor : Intel i3 or above

RAM: 8 GB Recommended

Storage : 10 GB Free Space

Software Requirement

window/linux

Burp suite

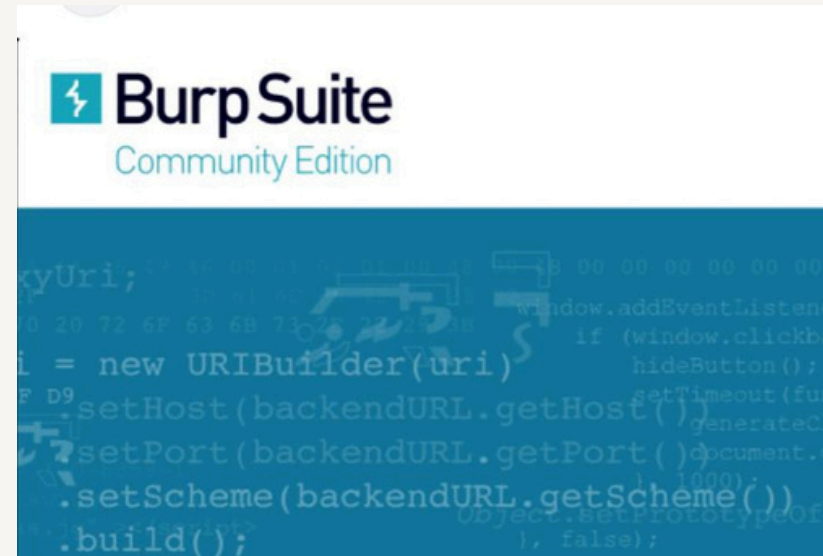
Google Chrome or Firefox

Testing Environment

Browser

Burp suite

Test Website



Lab Environment Setup

Step 1 : Launch Burp Suite

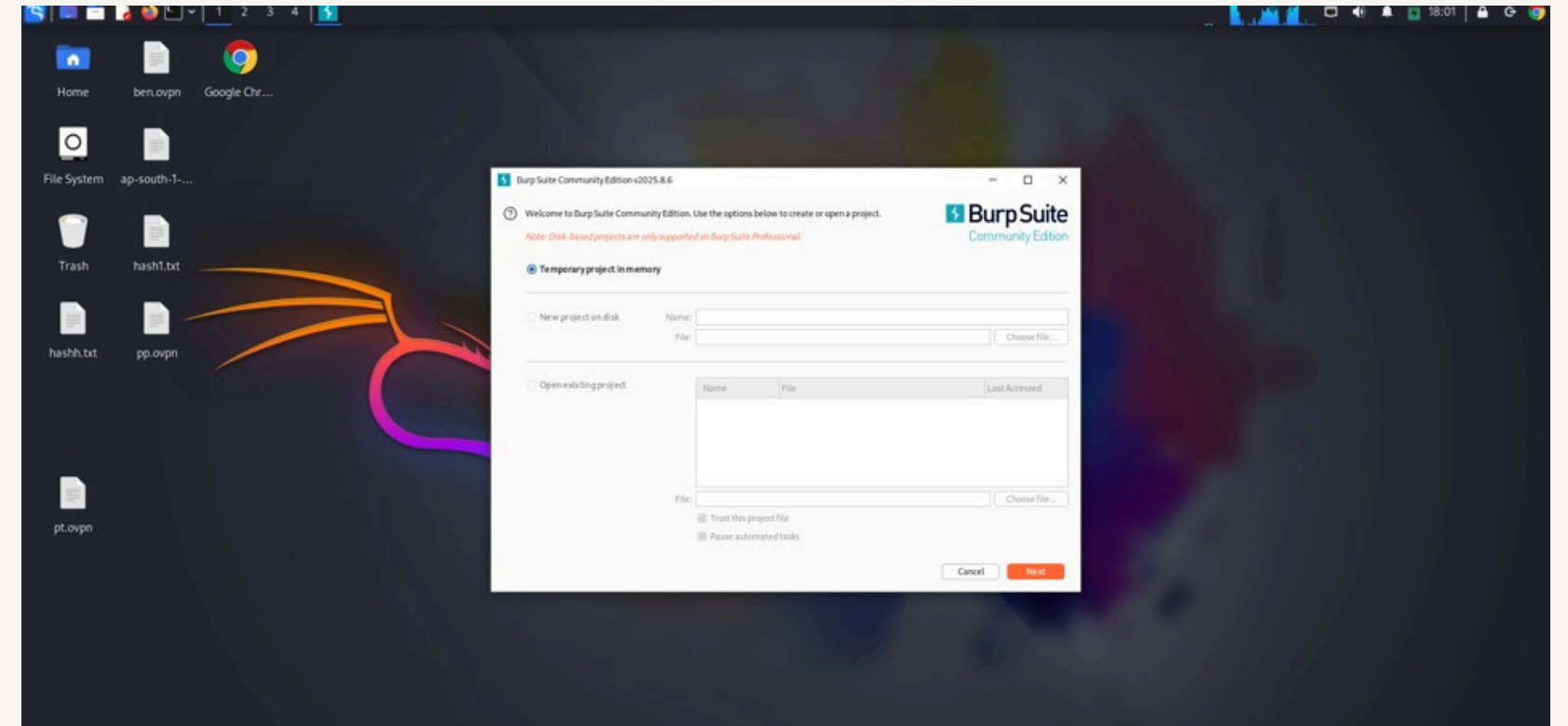
open kali linux

search for burp suite

Launch Burp suite community edition

Select Temporary Project

Click on start



Step 2 : Configure Browser Proxy

Burp suite works by intercepting browser traffic

Configuration Steps

open Firefox browser

Go to Network settings

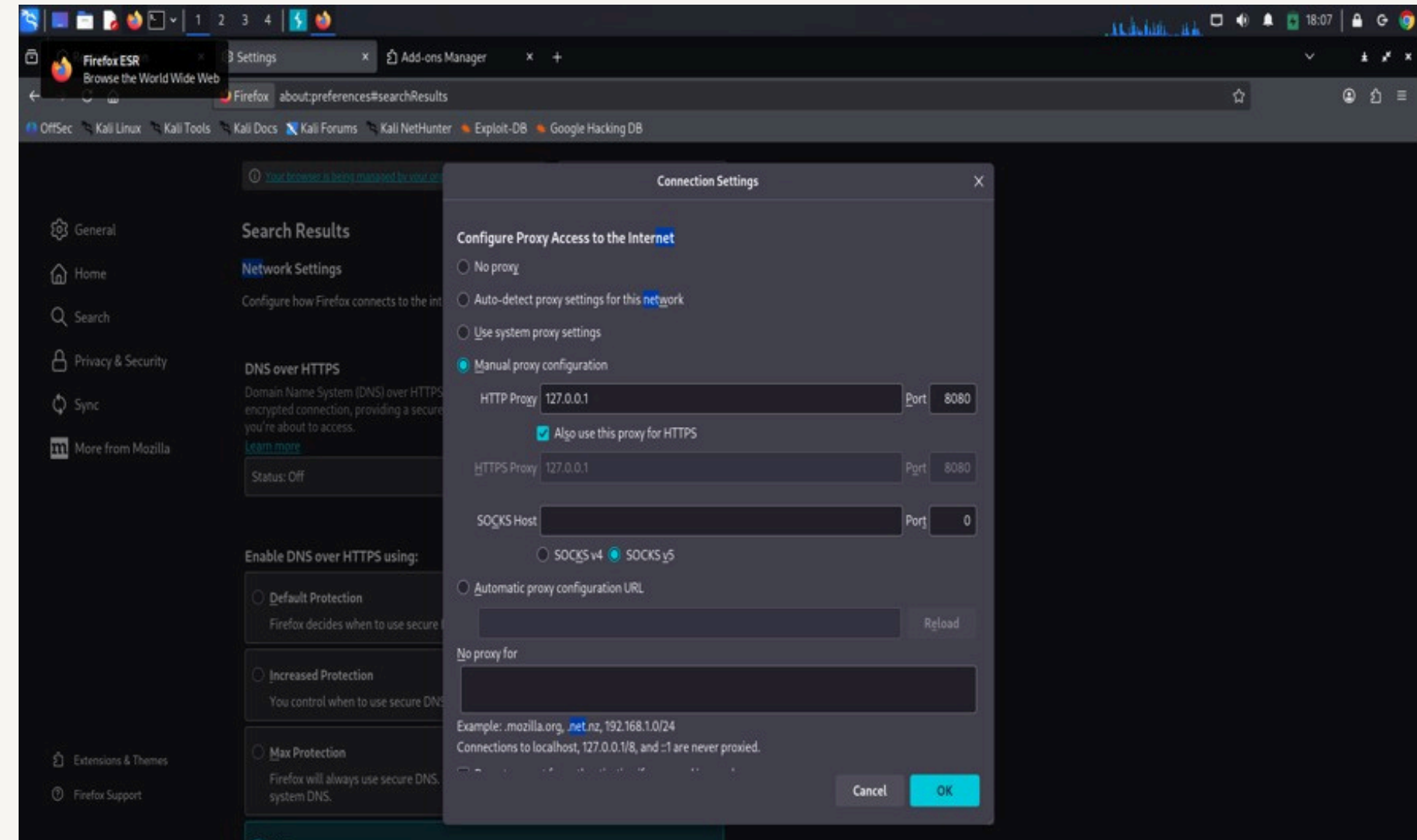
Select Manual Proxy Configuration

Enter:

IP Address:127.0.0.1

Port :8080

Save settings

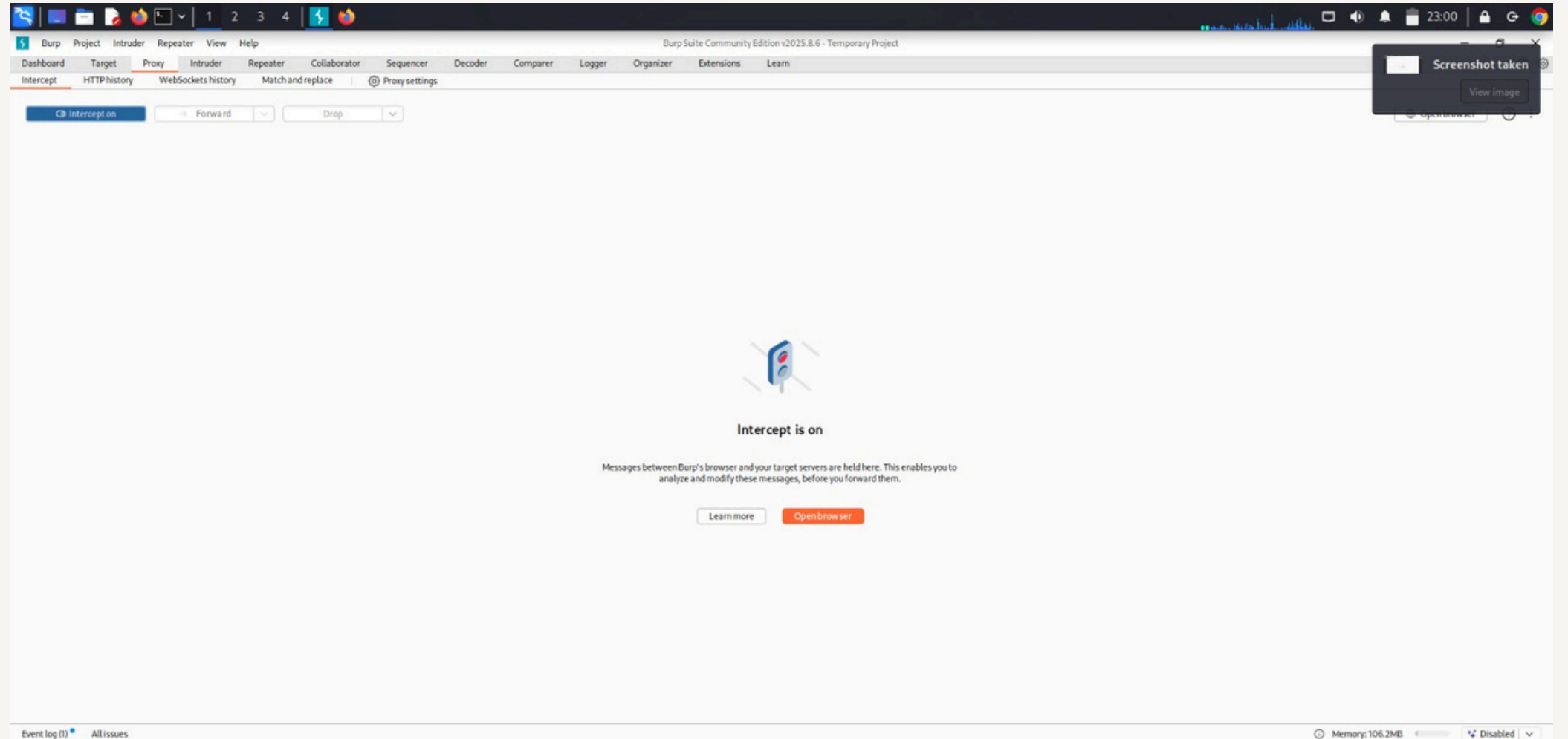


Step 3 : Enable interception

Go to Burp Suite

open Proxy Tab

Turn Intercept ON



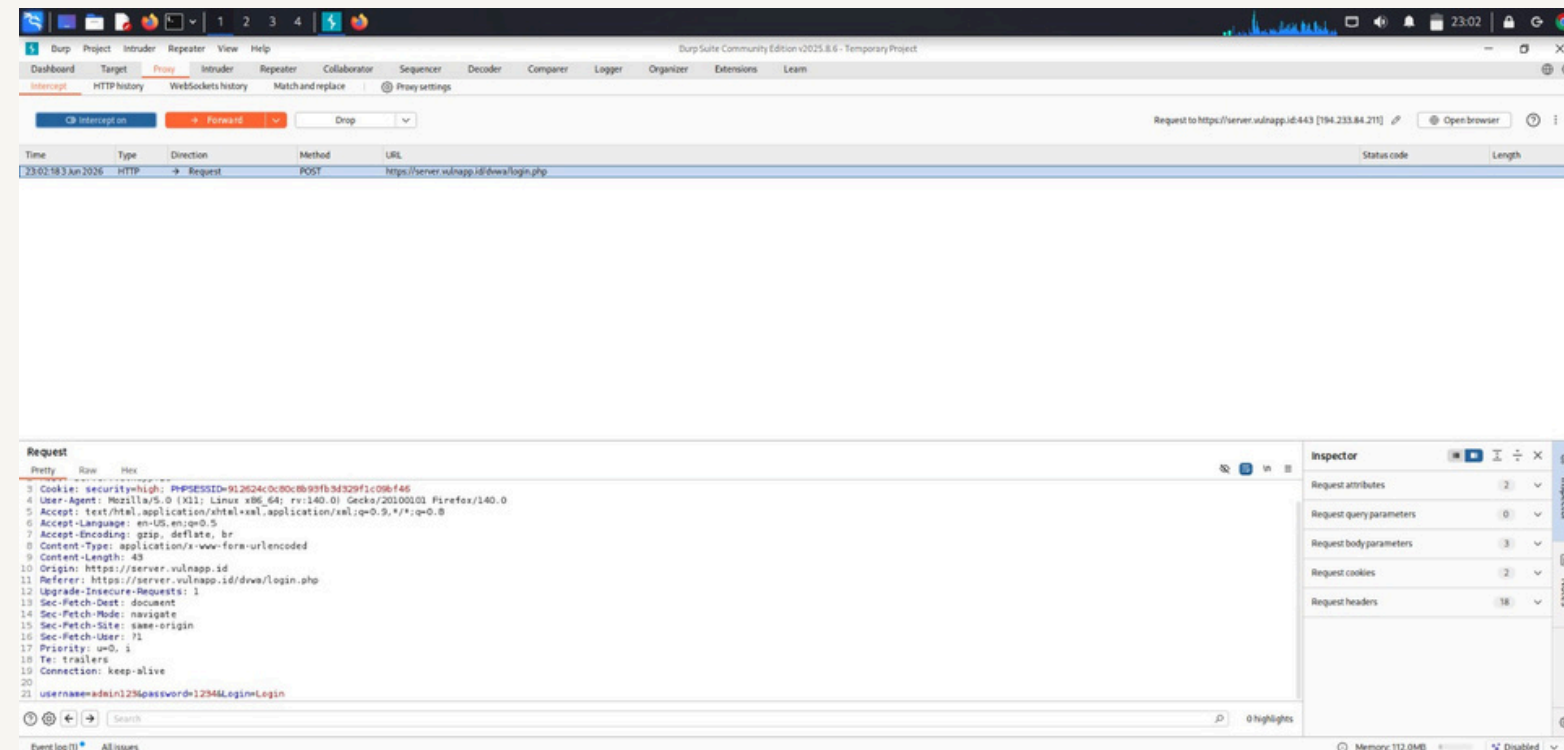
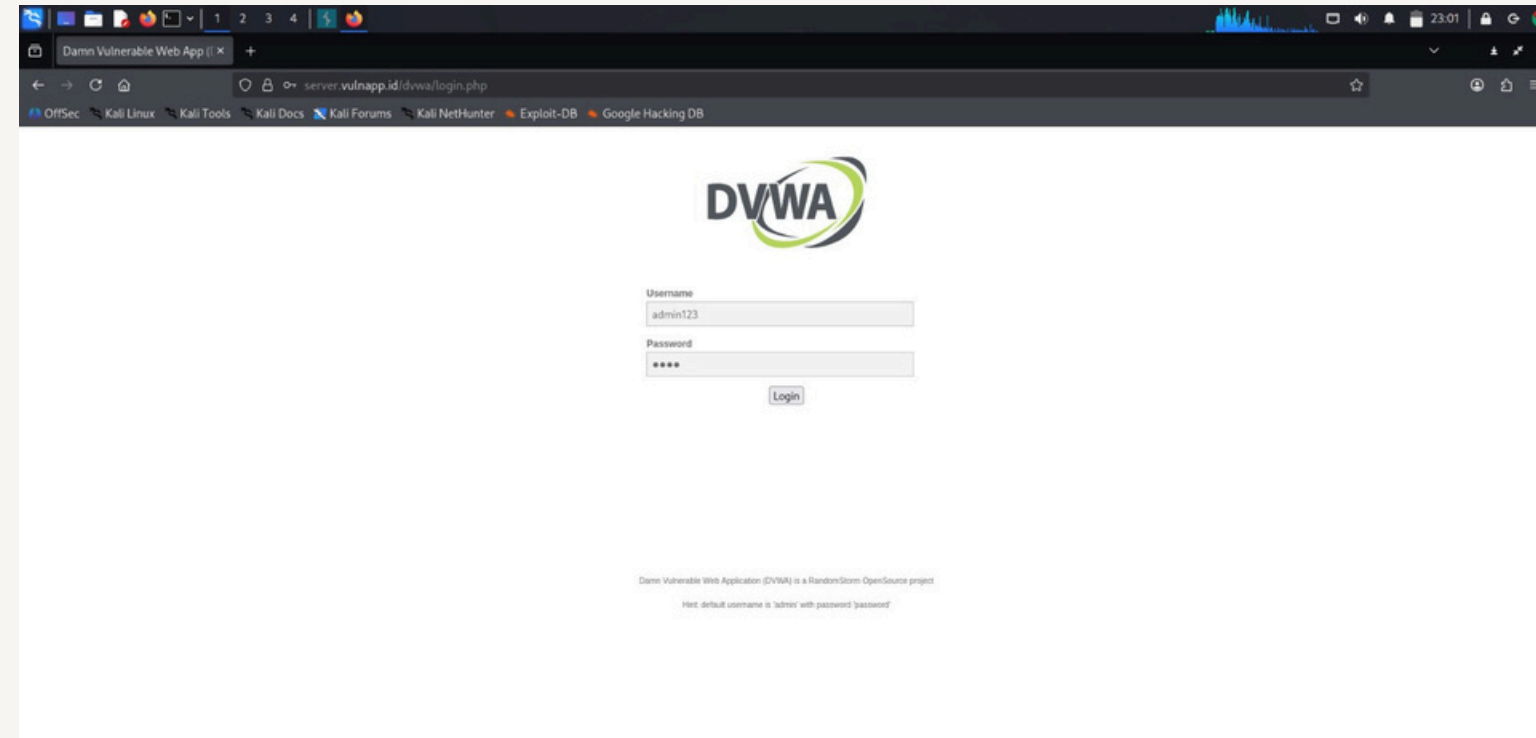
This allows Burp Suite to capture browser requests

Methodology

4.1 Traffic Interception

Process

1. open browser
2. Visit DVWA or test website
3. Enter login credentials
4. Burp suite intercept th request



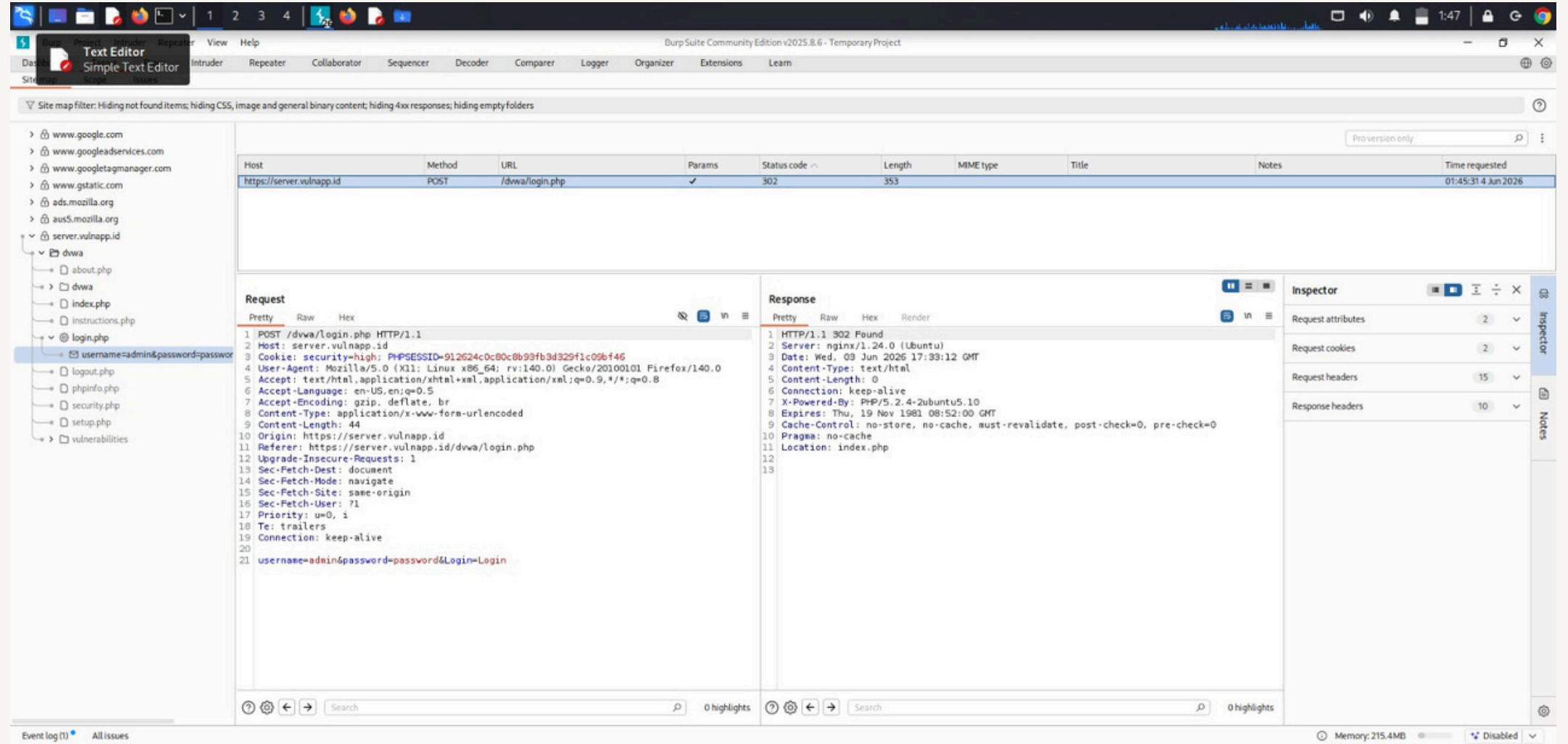
Burp suite capture HTTP requests sent from browser to server .This help Security ananlysts examine data transmitted between client and server

Request Analysis

Captured request contains

- Username and password
- Session Cookies
- Form parameters
- Headers

This information helps identify weak authentication mechanisms

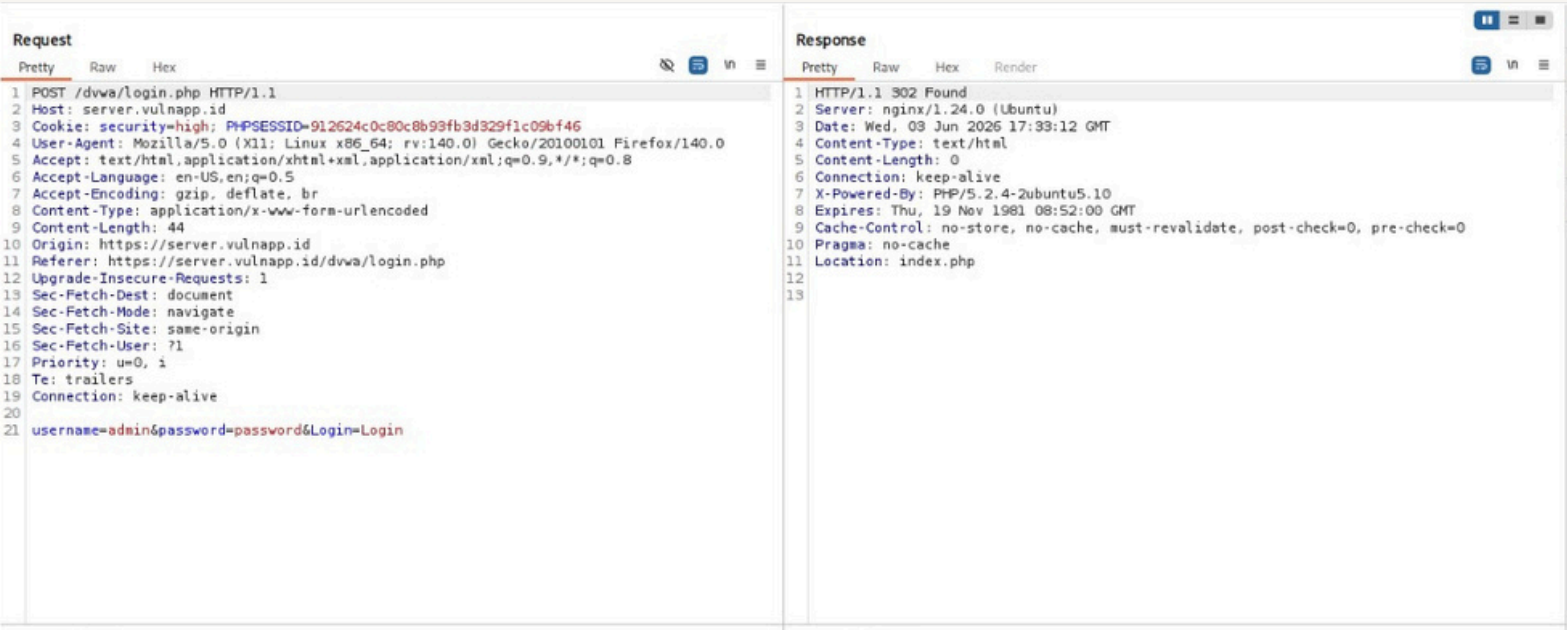
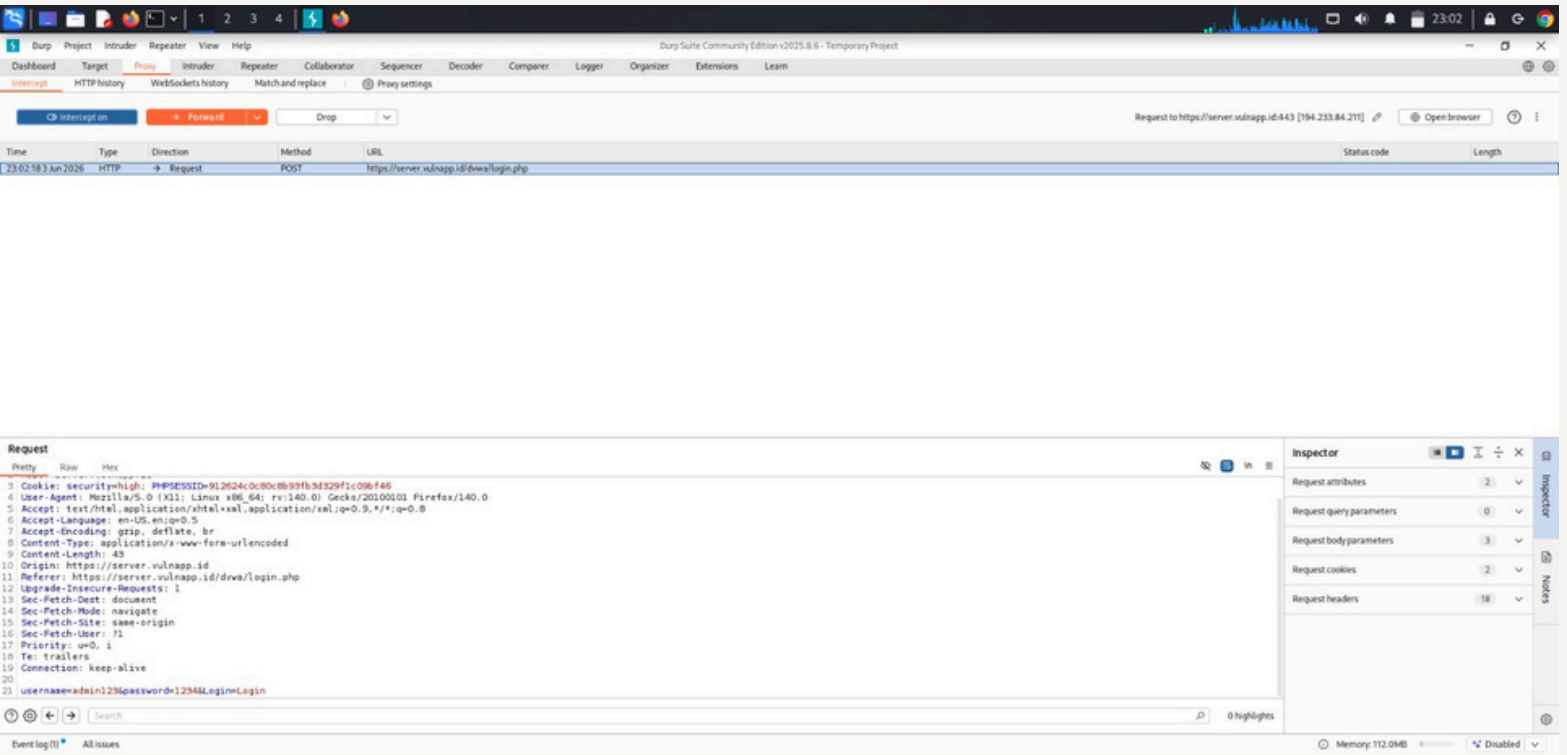


Request Modification

Burp Suite allow modification of request before sending them to the server

Example Testing

- Modify login credentials
- Change input parameters
- Test validation controls



SQL Injection Testing

SQL Injection is a vulnerability where attackers manipulate database queries.

Testing steps

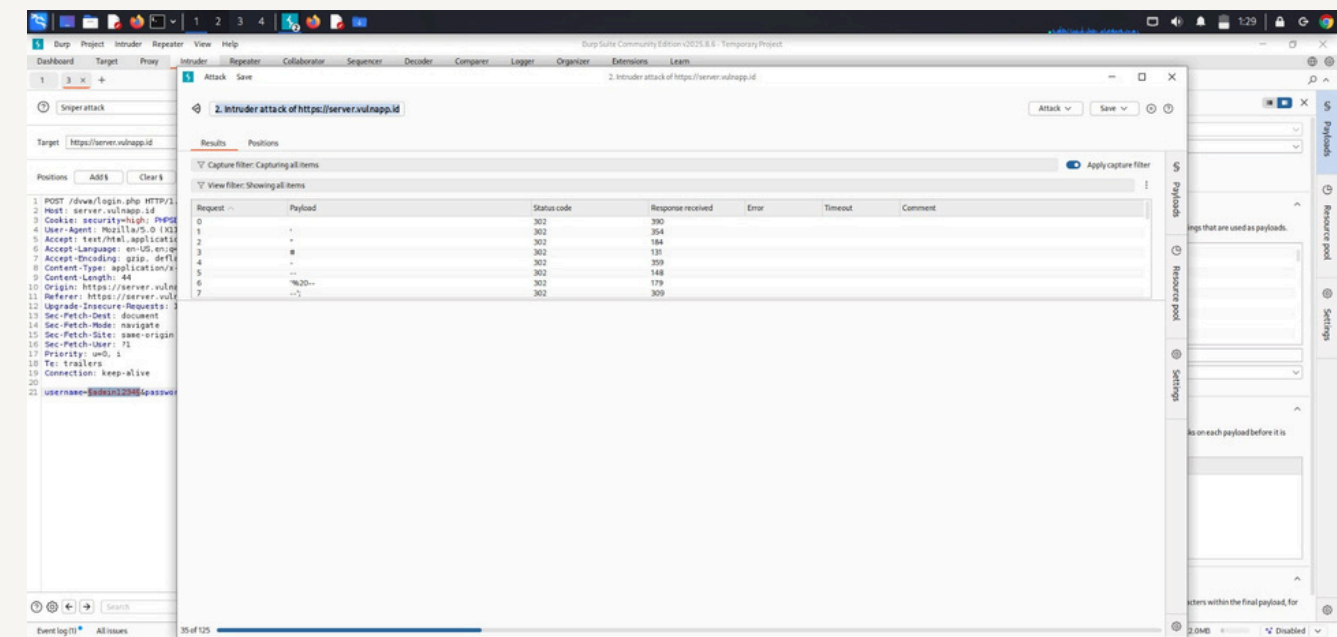
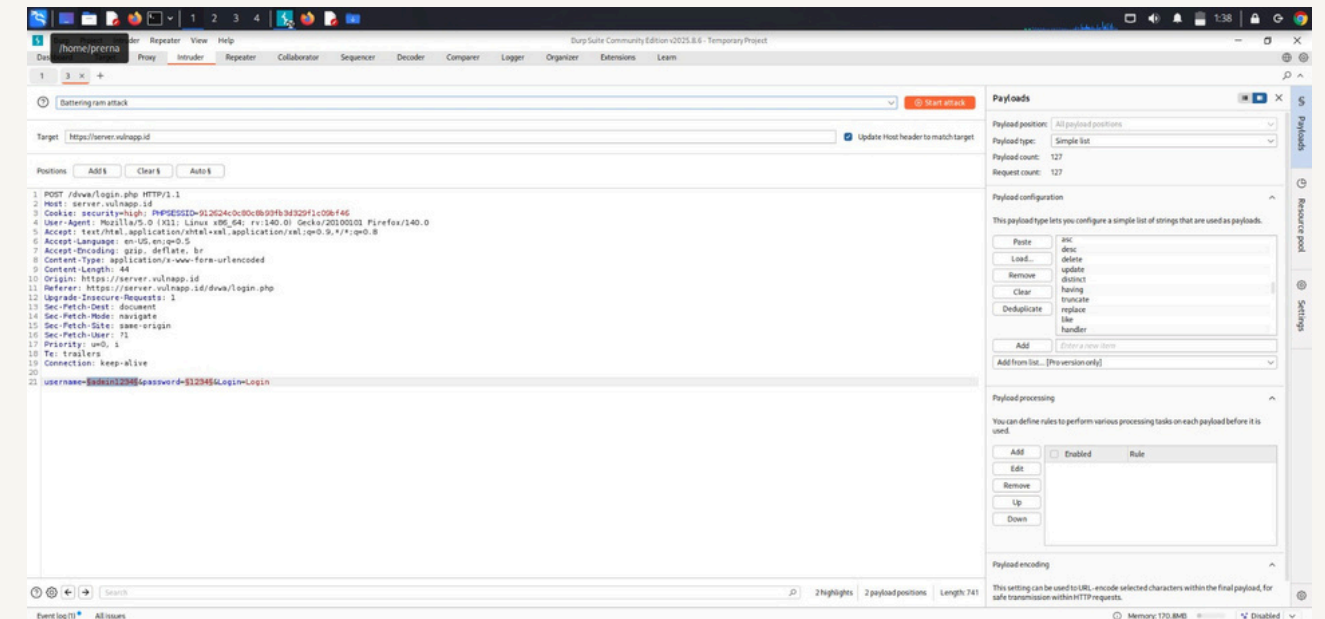
1. Intercept Login request

2. Modify username field

' OR '1'='1

3. Forward request

If authentication bypass occur, the application is Vulnerable



Session Handling Analysis

Burp suite helps analyze session cookies.

Weak session handling can lead to :

- Session hijacking
- Unauthorized account access

Request	
Pretty	Raw
1	GET /dvwa/index.php HTTP/1.1
2	Host: server.vulnapp.id
3	Cookie: security=high; PHPSESSID=912624c0c80c8b93fb3d329f1c09bf46
4	User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/14
5	Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
6	Accept-Language: en-US,en;q=0.5
7	Accept-Encoding: gzip, deflate, br
8	Referer: https://server.vulnapp.id/dvwa/login.php
9	Upgrade-Insecure-Requests: 1
10	...

Result Analysis

During testing, the following observations may occur:

Login authentication weakness.

Improper input validation

Exposure of session cookies

Possibility of injection attacks

These vulnerabilities indicate security flaws in web applications.

Damage Impact Analysis

If vulnerabilities are exploited ,consequences include:

Data Theft

Attackers may access confidential user information.

Account Takeover

Unauthorized user may gain control of accounts.

Website Defacement

Attacker may modify wesite content

Financial Loss

Sensitive transaction data may be compromised

Security Recommendations

To prevent web vulnerabilities

Implement Strong input Validation

Use parameterized queries

Enable secure session management

Encrypt sensitive data

Encrypt sensitive data

Implement Web Application Firewall (WAF)

Conclusion

Burp Suite is a powerful tool for web security testing. This project demonstrated how vulnerabilities in authentication, input validation, and session management can expose web applications to cyber attacks. Proper security practices are essential to prevent exploitation.